

Switching the AAD Reproduction to CICIoT2023

Dataset relevance, required changes, training cost, adversary realism and key statistics

Project: AAD/EIDS reproduction (Verma et al., IEEE TCE 2025) — FYP extension	Date: 7 October 2026	Current dataset: CSE-CIC-IDS2018 (2 files)
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1. Executive summary

2. Relevance of CICIoT2023

3. Changes to models and architecture

4. Training time estimates
5. Making the adversary realistic

6. Statistics you need to know

7. Pre-flight checks and blockers

8. Sources

1. Executive summary

- **CICIoT2023 is a strong fit.** It closes proposal gap #4 ("defenses motivated by IoT are validated on enterprise traffic") and is the dataset used by two works cited in the proposal (Yaras & Dener [2]; Yao et al. PATG [8]), so results become directly comparable.
- **The models barely change; the validator is the real rewrite.** All models are already parameterised by `n_features` (68 → 46). The validator's dependency rules are hard-coded to CICFlowMeter column names and must be rebuilt.
- **The integrality rule family largely disappears.** CICIoT2023 features are averages over packet windows, so counts are not whole numbers. The rule family that caught 100% of adversarial rows in Table V will no longer carry the defense — which makes the discriminator, EIDS and ablation (phases 5–7) meaningful for the first time.
- **Do not train on all 46.7M rows.** A capped stratified sample (~2.55M rows) trains in ~1.2 h for phase 2 and ~4–5 h end to end on the current laptop CPU. The full set would take ~22 h for phase 2 alone and does not fit in 15.7 GB RAM.
- **Realism comes from three levers:** evading only where evasion is functionally meaningful (Recon, Web, Brute Force, Spoofing), restricting perturbations to attacker-controllable features, and verifying at packet level using the released PCAPs and feature extractor.

2. Relevance of CICIoT2023 to the project

Aspect	CSE-CIC-IDS2018 (current)	CICIoT2023
Environment	Enterprise LAN	105 real IoT devices (67 IP-based, 38 Zigbee/Z-Wave)
Attacks	5 attack types (2 capture files)	33 attacks in 7 categories (34 labels incl. benign)
Rows	1.50M after cleaning	46,686,579
Features	68 (CICFlowMeter, per flow)	46 (DPKT, averaged over 10- or 100-packet windows)
Attack share	9.74%	97.65% — benign is the minority
Use in related work	AAD paper only	Yaras & Dener [2], Yao et al. PATG [8]
PCAPs + extractor code	Yes	Yes — enables packet-level realizability testing

Caveats to state in the write-up

- **Not medical.** The proposal introduction mentions IoMT. The same lab's **CICIoMT2024** uses a nearly identical feature pipeline, so a "train on CICIoT2023, test on CICIoMT2024" cross-dataset experiment is a natural addition.
- **97.6% of rows are floods (DDoS/DoS/Mirai).** Evading an IDS while still flooding is largely contradictory, so the classes worth evading are the small ones (Section 5).
- **No IPs, timestamps or flow IDs in the CSVs.** Grouped splitting by host and real temporal sequences for the LSTM are not possible from the CSVs; they would require going back to the PCAPs.

3. Changes to models and architecture

Phase	Change required	Effort
1. Data	New loader for 169 CSV files (~13 GB unzipped, ~2.7 GB zipped). New 34-entry <code>label_map</code> plus an 8-category field. Capped stratified sampling instead of loading 46.7M rows. Deduplication. Single-feature leakage probe. <code>log1p</code> before MinMax on heavy-tailed columns (Rate, Header_Length, flow_duration, Tot sum, IAT, Covariance, Weight) — otherwise MinMax compresses most rows near 0 and <code>eps = 0.1</code> stops meaning anything. Convert to Parquet once after ingest.	Medium
2. NIDS	Input becomes 46-dimensional automatically. Keep the 2-unit softmax binary head — the whole ART evasion chain depends on it. Enable <code>class_weight</code> or rebalance, as benign is the minority. Per-type recall reporting grows from 6 rows to 34 classes plus 8 categories (<code>evaluate.py</code>).	Small
3. Attacks	Source rows can finally be stratified across attack categories (currently 99.6% DoS-Hulk). <code>eps</code> means something different in log-scaled space, so the <code>eps</code> sweep must be re-run.	Small
4. Validator	Rewrite. <code>validator.py</code> lines 62–86 hard-code CICFlowMeter columns (<code>Fwd Pkt Len Min</code> , <code>Flow IAT Mean</code> , ...). See invariants below.	Large
5. Discriminator	Code unchanged. Add a leave-one-category-out protocol (e.g. train without Web, test on Web).	Small
6. EIDS	Unchanged apart from the larger adversarial-training pool.	—
7. Pipeline	Unchanged.	—
Tests	<code>test_validator.py</code> fixtures are built from CICFlowMeter violations and must be rewritten.	Small

Why the validator change matters beyond the code. Because CICIoT2023 features are window means, count features such as `syn_count` and `ack_count` are not integers. The integrality family — which alone caught 100% of adversarial rows in Table V — mostly stops applying. Table IX's finding that "the validator catches everything, so the discriminator and EIDS never see an adversarial flow" is very unlikely to hold. That makes phases 5–7 and the ablation genuinely informative, which strengthens the thesis.

Candidate replacement invariants

Let `analyzer.py` keep only the ones whose benign residuals are tight:

- $\text{Min} \leq \text{AVG} \leq \text{Max}$ (packet length)
- $\text{Tot sum} \approx \text{AVG} \times \text{Number}$
- $\text{Srate}, \text{Drate} \leq \text{Rate}$
- Flag rates and protocol indicators within $[0, 1]$

- HTTP / HTTPS / SSH / Telnet / SMTP / IRC $\leq$ TCP and DNS / DHCP $\leq$ UDP
- TCP + UDP + ICMP $\leq$ 1
- Flag counts bounded by the TCP share
- Protocol Type consistent with the protocol indicators

Range and distribution (median/MAD z-score) rule families carry over unchanged.

4. Training time estimates

Hardware: Intel i5-12450H (8 cores / 12 threads), 15.7 GB RAM, CPU-only TensorFlow 2.13, 36.5 GB free on D:.

Estimates scale the measured per-row times from the current run (CNN 1,237 s and LSTM 2,486 s for 30 epochs on 1.05M rows $\times$ 68 features) by 46/68 for the shorter input. All figures assume 30 epochs; early stopping usually finishes sooner.

Training rows (70% split)	MLP	CNN	LSTM	Phase 2 total
1.8M — recommended (all benign + $\leq$ 60k per attack class $\rightarrow$ 2.55M total)	~2 min	~24 min	~48 min	~1.2 h
3.5M (cap 150k per class)	~5 min	~47 min	~1.6 h	~2.4 h
32.7M (entire dataset)	~42 min	~7.3 h	~14.6 h	~22 h — does not fit in RAM

End-to-end at the recommended size (~4–5 h compute)

Phase	Estimated time	Notes
1. Data	20–30 min once; 2–5 min after	One-time CSV ingest, then Parquet
2. Baselines	~1.2 h	LSTM dominates
3. AE generation	~15 min	Current run took ~11 min
4. Analyzer + validator	a few minutes	
5. Discriminator	~1 min	8 RF fits
6. EIDS	1.5–2 h + ~15 min	Retraining + adaptive evaluation
7. Pipeline	~5 min	

Development effort: roughly **3–5 days**, most of it in the loader and the validator rewrite.

Do not train on all 46.7M rows. The feature matrix alone is ~8 GB as float32 and pandas roughly doubles that during loading. The extra rows are near-identical flood windows that add no information. Unzipping on C: (229 GB free) keeps D: comfortable.

5. Making the adversary realistic

a. Only evade where evasion is functionally meaningful

Build the phase-3 source set per category:

Category	Realistic evasion?	Attacker's constraint
DDoS / DoS / Mirai (95.8% of rows)	Weak	Rate must stay above a floor. Report the throughput lost — cutting the rate by 90% is mitigation, not evasion.
Recon (scans, ping sweep)	Yes	Can slow down, randomise TTL, pad packets. Cost = longer scan time.
Web (SQLi, XSS, CmdInj, Upload, Backdoor, Browser Hijack)	Yes	HTTP and TCP fixed; payload can grow but not shrink below the exploit.
Brute force	Yes	Throttling attempts is cheap.
Spoofing (ARP, DNS)	Partly	Protocol fixed; timing adjustable.

b. Restrict perturbations to what an attacker can change

Mutability class	Features
Immutable	Protocol Type; HTTP, HTTPS, DNS, Telnet, SMTP, SSH, IRC, TCP, UDP, DHCP, ARP, ICMP, IPv, LLC indicators
Increase-only	Packet sizes via padding (Min, Max, AVG, Tot size); Header_Length (TCP options); IAT and flow_duration (added delay)
Decrease-only (at a cost)	Rate, Srate
Freely settable	Duration (TTL); PSH / URG / ECE / CWR flags on crafted packets
Derived — recompute, never perturb directly	Std, Variance, Covariance, Magnitue, Radius, Weight, Number, *_count , Tot sum

c. Prove realizability at packet level

CICIoT2023 ships the original PCAPs and its DPKT-based feature extractor. Apply the perturbation to real packets (Scapy: padding, delays, TTL and flag changes), re-run the extractor, and feed the resulting features to AAD. This is end-to-end realizability — proposal gap #3, and exactly what Hasan et al. [7] and Apruzzese et al. [6] call for.

d. Defense-aware adaptive attacker (phase 8)

- **Attack the ensemble jointly:** maximise the *minimum* benign margin across MLP, CNN and LSTM, rather than attacking one member.
- **Get past the validator:** add a penalty on the validator's distribution z-score and project onto the constraint set at every step.
- **Get past the RF discriminator:** distil the forest into a small differentiable MLP and use it as a gradient surrogate, or use BPDA (Athalye et al.).

e. Weaker-knowledge threat tiers

- **Grey-box:** transfer attacks from a surrogate trained on a disjoint split or on CICIoMT2024.
- **Black-box:** HopSkipJump (available in ART), reporting the number of queries used.

f. Zero-day test at the IDS level

Train the NIDS with a whole category held out (Web or Recon) and test on it, so the evaluation includes attacks the defense has never seen.

6. Statistics you need to know

Category	Labels	Rows	Share
DDoS	12	33,984,560	72.79%
DoS	4	8,090,738	17.33%
Mirai	3	2,634,124	5.64%
Benign	1	1,098,195	2.35%
Spoofing	2	486,504	1.04%
Recon	5	354,565	0.76%
Web	6	24,829	0.05%
Brute force	1	13,064	0.03%
Total	34	46,686,579	100%

Recon here includes VulnerabilityScan. The CIC dataset page lists 11 DDoS attacks, but the CSV carries 12 DDoS labels (DDoS-ICMP_Fragmentation appears as its own label). Category totals sum exactly to the published 46,686,579 rows.

- **Imbalance:** 5,751 : 1, from DDoS-ICMP_Flood (7,200,504 rows) to Uploading_Attack (1,252 rows).
- **Trivial baseline:** a model that always predicts "attack" scores **97.65% accuracy**. Report balanced accuracy, benign FPR and per-category recall — never accuracy alone.
- **Rare classes become measurable:** SQL injection gets ~787 test rows instead of 7; Web + Brute Force together give ~5,700 test rows instead of 81.
- **Literature ceiling:** binary accuracy of 99%+ is routine on this dataset. Expect the same high-accuracy pattern the README already criticises, so keep the per-type recall table alongside headline figures.
- **Recommended sample size:** ~2.55M rows (all benign + ≤60k per attack class), giving ~1.78M train / ~382k val / ~382k test at 70/15/15, with ~43% benign.

Largest and smallest classes

Class	Rows	Class	Rows
DDoS-ICMP_Flood	7,200,504	DictionaryBruteForce	13,064
DDoS-UDP_Flood	5,412,287	BrowserHijacking	5,859
DDoS-TCP_Flood	4,497,667	CommandInjection	5,409
DDoS-PSHACK_Flood	4,094,755	SqlInjection	5,245
DDoS-SYN_Flood	4,059,190	XSS	3,846
DDoS-RSTFINFlood	4,045,285	Backdoor_Malware	3,218
DDoS-SynonymousIP_Flood	3,598,138	Recon-PingSweep	2,262
DoS-UDP_Flood	3,318,595	Uploading_Attack	1,252

7. Pre-flight checks and blockers

Blocker — broken environment. The project `.venv` cannot start: its base interpreter (Python 3.10, `...\Programs\Python\Python310`) is no longer installed. Reinstall Python 3.10.11 or rebuild the venv before running anything. TensorFlow 2.13 still requires Python 3.10 or 3.11.

Run these probes in phase 1 before trusting any results:

1. **Exact-duplicate rate per class.** Flood windows repeat heavily; on CIC-IDS2018 the SlowHTTPTest class kept only 0.04% of rows after deduplication.
2. **Single-feature leakage.** Train a shallow tree on each feature alone; any feature giving >99% binary accuracy is suspect. IAT is reportedly very dominant on this dataset, and Number / Weight will leak the label if the window size (10 vs 100 packets) differs by class.
3. **Test-vs-train overlap** — the same byte-identical-row probe already run for CIC-IDS2018.

8. Sources

- Canadian Institute for Cybersecurity — CICIOT2023 dataset page: <https://www.unb.ca/cic/datasets/iotdataset-2023.html>
- Neto et al., "CICIOT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment," *Sensors* 23(13), 5941, 2023: <https://www.mdpi.com/1424-8220/23/13/5941>
- nickjeffrey, CICIOT2023 preprocessing notes: https://github.com/nickjeffrey/ensemble_learning/blob/main/CIC_IOT_Dataset2023_dataset_preprocessing.md
- EmergentMind, CICIOT2023 topic summary: <https://www.emergentmind.com/topics/ciciot2023-dataset>
- Kaggle mirror: <https://www.kaggle.com/datasets/madhavmalhotra/unb-cic-iot-dataset>
- Project materials: `aad/README.md`, `aad/config/*.yaml`, `aad/src/aad/`, `aad/artifacts/reports/`, `Proposal.pdf`